

Problem Statement Document AI Study & Exam Management Platform

The increasing dependency on digital learning resources has created a challenge for students in managing study materials, revision schedules, and exam preparation efficiently. Students often spend excessive time organizing notes, creating summaries, preparing quizzes, and revising important topics manually. Existing educational platforms generally provide static content without personalization, automation, or AI-driven assistance.

Problem Statement

Develop an AI-powered Study and Exam Management Platform that helps students streamline their academic preparation by automating note summarization, quiz generation, revision planning, and performance tracking. The system should integrate Artificial Intelligence APIs, full-stack architecture, database management, authentication systems, file processing, and document automation to provide an intelligent and scalable academic assistant.

Project Objectives Provide centralized management of study materials. Generate AI-powered summaries and quizzes from uploaded content. Create automated revision schedules and study analytics. Allow students to track progress and academic performance. Generate downloadable reports and revision PDFs. Improve productivity and reduce manual academic workload.

Scope of the Project User Authentication and Authorization PDF/Document Upload and Parsing AI-based Content Analysis Quiz and MCQ Generation Dashboard and Analytics Revision Planner and Notifications PDF Report Generation Cloud Storage Integration

Core Features Secure login/signup using JWT or OAuth. Upload syllabus, notes, and study materials. Automatic extraction and summarization of study content. AI-generated MCQs, flashcards, and short notes. Personalized study dashboard with analytics. Progress tracking and revision reminders. Downloadable study reports in PDF format. Email notifications for schedules and reminders.

Recommended Technology Stack

Layer	Technology
Frontend	Next.js / React
Backend	FastAPI / Node.js
Database	PostgreSQL
Authentication	JWT / OAuth
AI Integration	Gemini API / OpenAI API
Storage	Cloudinary / AWS S3
Deployment	Vercel / Railway / Docker

Future Enhancements AI chatbot for doubt solving using RAG architecture. Voice-based interaction and lecture transcription. Collaborative study groups and peer learning. Mobile application support. Advanced analytics using machine learning.

Conclusion

The AI Study & Exam Management Platform aims to modernize academic preparation using Artificial Intelligence and

automation. The project demonstrates practical implementation of full-stack development, API integration, database management, authentication systems, and automated document generation while solving a real-world educational problem.